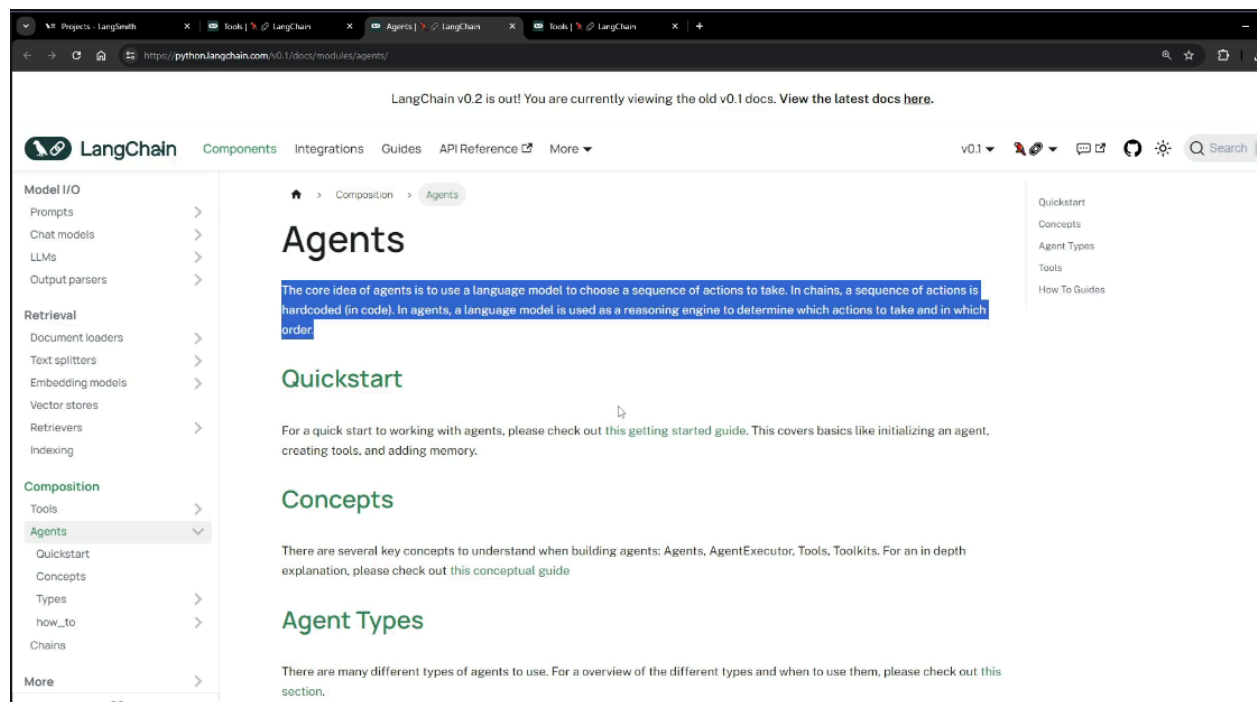
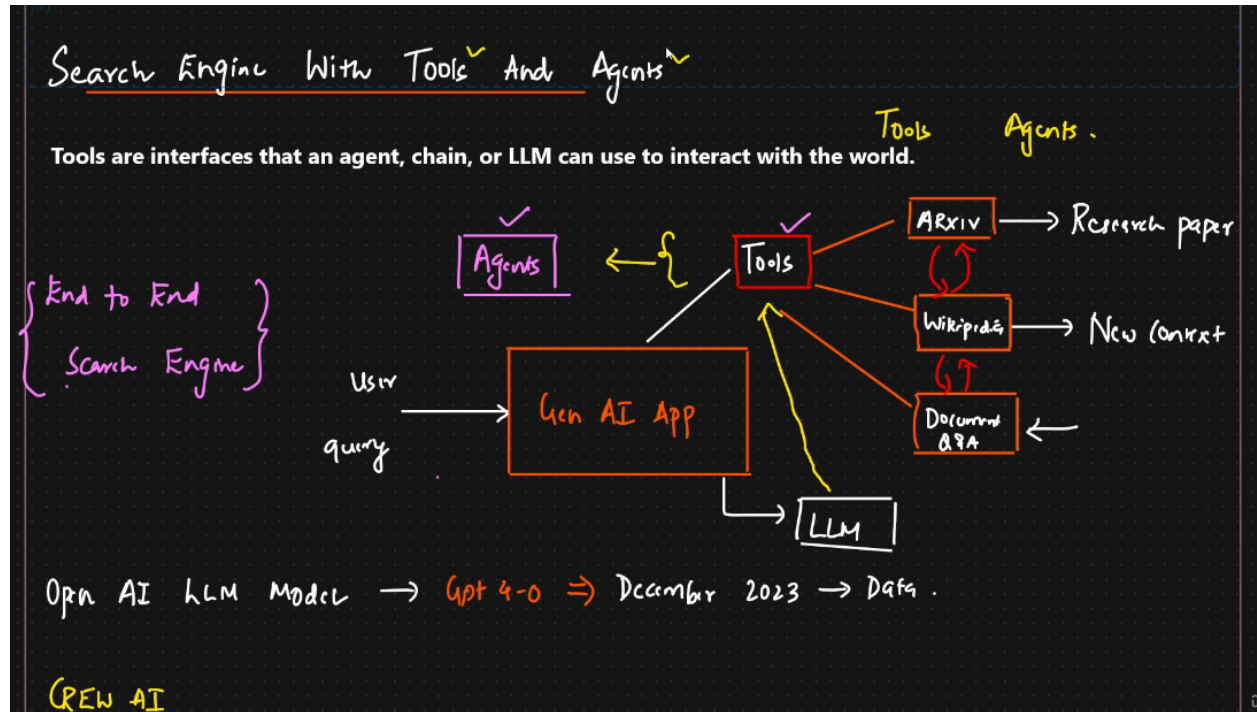


if we need to interact with all tools in the below picture then we need agents.

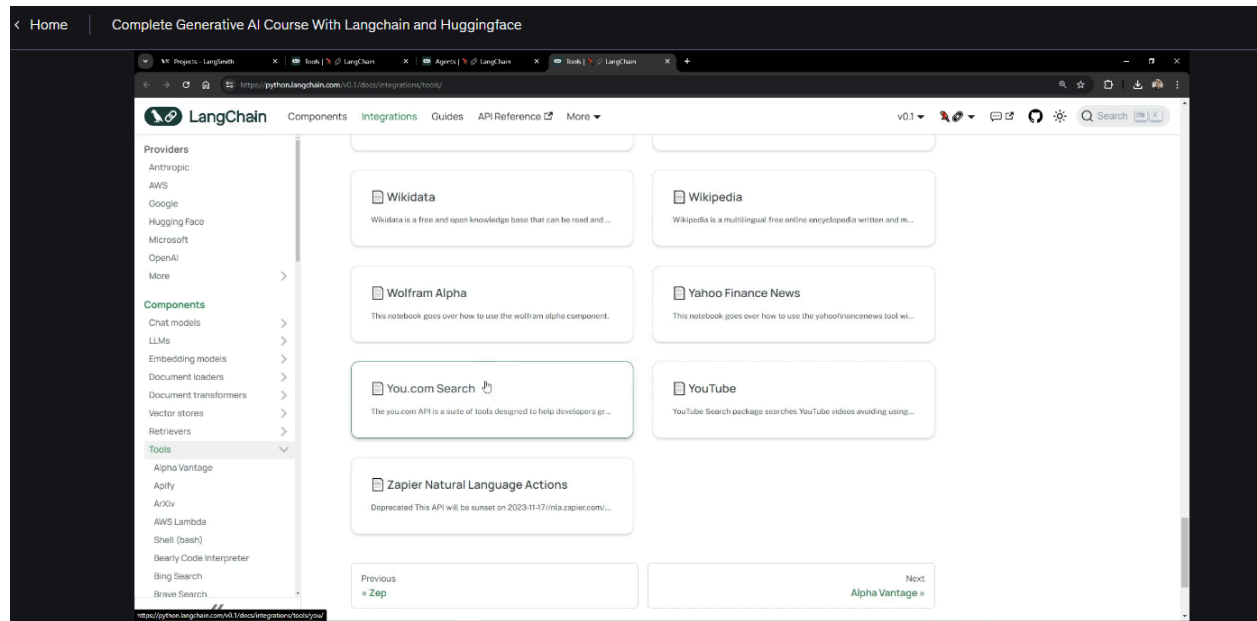
langchain provides entire ecosystem

There are also other libraries called CREW AI, here also we use multiple tools and agents.

in this project we will also make interaction happen with each tools using agents.



Langchain has lot of tools



Refer to the file C:\Users\harig\Downloads\Machine learning sections\Section 33 Search engine with Langchain tools and agents\5-Search+Engine\5-Search Engine\tools_agents.ipynb, this file will tell how to use the tools in python.

here multiple agents are there, that we can create

```
tools_agents.ipynb × app.py requirements.txt
5-Search Engine > tools_agents.ipynb > ...
+ Code + Markdown | ▶ Run All ↺ Restart ≡ Clear All Outputs | [L] Variables ≡ Outline ... venv (Python
from langchain import hub
prompt=hub.pull("hwchase17/openai-f
prompt.messages

[13] ✓ 1.5s

... [SystemMessagePromptTemplate(prompt=Pro
MessagesPlaceholder(variable_name='ch
HumanMessagePromptTemplate(prompt=Pro
MessagesPlaceholder(variable_name='agi

Agents
from langchain.agents import create
```

LangChain v0.2 is out! You are currently viewing the old v0.1 docs. [View the latest docs here.](#)

 LangChain

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OpenAI tools

Newer **OpenAI models have been** fine-tuned to detect when **one or more** function(s) should be called and respond with the inputs that should be passed to the function(s). In an API call, you can describe functions and have the model intelligently choose to output a JSON object containing arguments to call these functions. The goal of the OpenAI tools APIs is to more reliably return valid and useful function calls than what can be done using a generic text completion or chat API.

OpenAI termed the capability to invoke a **single** function as **functions**, and the capability to invoke **one or more** functions as **tools**.

TIP

In the OpenAI Chat API, **functions** are now considered a legacy options that is deprecated in favor of **tools**.

If you're creating agents using OpenAI models, you should be using this OpenAI Tools agent rather than the OpenAI functions agent.

Using **tools** allows the model to request that more than one function will be called upon when appropriate.

In some situations, this can help significantly reduce the time that it takes an agent to achieve its goal.

See

- [OpenAI chat create](#)
- [OpenAI function calling](#)

Initialize Tools

Create Agent

Run Agent

Using with chat history

```

if prompt:=st.chat_input(placeholder="What is machine learning?"):
    st.session_state.messages.append({"role":"user","content":prompt})
    st.chat_message("user").write(prompt)

    llm=ChatGroq(groq_api_key=api_key,model_name="Llama3-8b-8192",streaming=True)
    tools=[search,arxiv,wiki]

    serach_agent=initialize_agent([tools,llm,agent=AgentType.Ze])

```

☐ ZERO_SHOT_REACT_DESCRIPTION
☐ CHAT_ZERO_SHOT_REACT_DESCRIPTION
☐ .capitalize capitalize
☐ STRUCTURED_CHAT_ZERO_SHOT_REACT_DESCRIPTION

zero shot react descripion, doesn't depend on chat history

Chat_zero_shot_react_description depends on chat history for the agent response.